

Erdős Problem #201

EP-201

L1 - TRACTABLE

OPEN

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $G_k(N)$ be such that any set of N integers contains a subset of size at least $G_k(N)$ which does not contain a k -term arithmetic progression. Determine the size of $G_k(N)$. How does it relate to $R_k(N)$, the size of the largest subset of $\{1, \dots, N\}$ without a k -term arithmetic progression? Is it true that $\lim_{N \rightarrow \infty} \frac{R_3(N)}{G_3(N)} = 1$?

Source and attribution. Statement reproduced from UnsolvedMath, whose official dataset is released under the CC BY 4.0 license.
Archive: <https://www.unsignedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>